

# The shortest relations of planar triangle reflection groups

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## Abstract

Let  $\tau$  be a Euclidean triangle and  $G_\tau$  the group generated by the three reflections in its sides. We determine the shortest relations satisfied by these generators for every  $\tau$ : there are exactly 33 of them, each equating two reduced words of length 11, and for all  $\tau$  outside a proper Zariski-closed set of shapes they are the only coincidences in the ball of radius 11. The associated growth sequence begins 1, 3, 6, 12, 24, 48, 96, 192, 384, 768, 1536, 3039, 6012. We complement this with the word metric of the translation subgroup: its elements are signed configurations of lamps on a hexagonal site lattice, and we identify them with the finitely supported integer flows on the honeycomb Cayley graph of the point group  $\mathbb{Z}^2 \rtimes C_2$ . For all shapes outside countably many proper subvarieties, the word length of a configuration equals the  $\ell^1$ -norm of its flow plus twice the least number of edges connecting the support of the flow to the base vertex. The cancellation and single-lamp laws follow, and the formula reproduces the exact census of translations through length 18, element by element. The shape space is stratified: on the lines where one angle equals  $\pi/m$  the relation onset moves to depth 6, 8, 9, 10, 10 for  $m = 4, 6, 3, 2, 8$  and is the universal one for the remaining  $m \leq 12$ , so that the crystallographic orders are not the distinguished ones. We prove that each divisor  $d \geq 2$  of  $m$  contributes  $d^2 - 1$  relations at depth  $m + m/d$ , which for even  $m$  gives  $(m/2)^2 - 1$  relations at depth  $m + 2$ .

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# 1 Introduction

Let  $\tau \subset \mathbb{R}^2$  be a nondegenerate Euclidean triangle and let  $r_0, r_1, r_2$  be the reflections in the lines containing its three sides. They generate a group  $G_\tau \leq \text{Isom}(\mathbb{R}^2)$ , the *triangle reflection group* of  $\tau$ . When the angles of  $\tau$  are of the form  $\pi/n_i$  the triangle tiles the plane and  $G_\tau$  is a Euclidean reflection group with the classical finite presentation. For all other shapes the group is not crystallographic, and Isola and Piergallini [1] showed that for a *generic* triangle, meaning one whose edge lengths are algebraically independent, the translation subgroup  $T_\tau$  is free abelian of infinite rank, the orientation preserving subgroup is free metabelian of rank two, and  $G_\tau$  admits no finite presentation. Their work determines the relations of  $G_\tau$  as a group. It does not address the word metric.

The present paper takes up the metric question. Write  $F = C_2 * C_2 * C_2 = \langle x_0, x_1, x_2 \mid x_i^2 \rangle$  and let  $\rho_\tau: F \rightarrow G_\tau$  send  $x_i$  to  $r_i$ . Elements of  $F$  are reduced words, and the sphere of radius  $d$  in  $F$  has  $3 \cdot 2^{d-1}$  elements for  $d \geq 1$ . The question is where, and by how much, the image sphere is smaller.

**Theorem 1.1** (Census). *There is an explicit list of 33 pairs  $(w_i, w'_i)$  of reduced words of length 11, pairwise distinct in  $F$ , such that*

- (i)  $\rho_\tau(w_i) = \rho_\tau(w'_i)$  for every nondegenerate Euclidean triangle  $\tau$  and every  $i$ ;
- (ii) for  $\tau$  outside a proper Zariski-closed subset of the shape space, these are the only coincidences of  $\rho_\tau$  on the ball of radius 11, and on the ball of radius 12 the only further coincidences are the 132 described in Remark 2.1.

Consequently the growth sequence  $u_d(\tau) = \#\{g \in G_\tau : \ell(g) = d\}$  of the word metric of a generic triangle begins

$$1, 3, 6, 12, 24, 48, 96, 192, 384, 768, 1536, 3039, 6012,$$

its first deviation from  $3 \cdot 2^{d-1}$  occurring at  $d = 11$  with deficit 33, and the shortest elements of  $\ker \rho_\tau$  common to all triangles have length exactly 22.

The second half of the paper concerns  $T_\tau$ . By [1, Thm. 4.1] it is free abelian on the conjugacy class of  $t_1 = (r_0 r_1 r_2)^2$ , so its elements are finitely supported integer vectors on a lattice of *sites*; we call a nonzero entry a *lamp*. Conjugation by the generators acts on sites through three affine involutions whose pairwise products are translations by the six vectors of a hexagonal

lattice (§3). In §5 we identify  $T_\tau$ , for generic shapes, with the group of finitely supported integer flows on the Cayley graph  $X$  of the point group  $\mathbb{Z}^2 \rtimes C_2$ , a cubic graph whose hexagonal faces correspond to the sites, and determine the metric completely:

**Theorem 1.2** (see Theorem 5.7). *For flow-generic  $\tau$  (Definition 5.4) and every reduced configuration  $c$ ,*

$$\ell(c) = \sum_{\{u,v\}} |c_u - c_v| + 2 \operatorname{st}(c),$$

*the sum over unordered hex-adjacent pairs of sites with  $c$  extended by zero, where  $\operatorname{st}(c)$  is the least number of edges of  $X$  whose addition connects the support of the flow of  $c$  together with the base vertex.*

The first summand is the  $\ell^1$ -norm of the flow of  $c$ ; the second is a Steiner term. The cancellation rule for lamps, the single-lamp costs of §4, and the exact census of translations by word length all follow (§5).

Section 6 records what happens on the non-generic strata. If one angle is exactly  $\pi/m$  and the shape is otherwise generic, then the reference group is the Coxeter group  $\langle x_i \mid x_i^2, (x_0 x_1)^m \rangle$ , whose growth series we compute in closed form, and the first deficit occurs at a depth  $d^*(m)$  which, for  $m \leq 12$ , is smaller than the universal 11 exactly when  $m \in \{2, 3, 4, 6, 8\}$ . The crystallographic orders are therefore not the ones admitting early relations, correcting the expectation that the crystallographic restriction governs the stratification: what governs it is the order of the rotations that the stratum creates, through the divisors of  $m$ .

## 2 The census theorem

Every triangle is similar to one with vertices  $(0,0)$ ,  $(1,0)$ ,  $(p,q)$  with  $q \neq 0$ , and similarity conjugates  $\rho_\tau$ , so all coincidence data depends only on  $(p,q)$ . Over  $\mathbb{Q}(p,q)$  the three reflections have matrix entries that are rational functions whose denominators are the squared side lengths, and these do not vanish for nondegenerate  $\tau$ .

*Proof of Theorem 1.1.* For (i): for each of the 33 pairs the identity  $\rho(w_i) = \rho(w'_i)$  is verified by exact arithmetic in  $\mathbb{Q}(p,q)$ . An identity of rational functions holds at every nondegenerate specialization, acute or obtuse.

For (ii): at the rational witness  $\tau_0$  with third vertex  $(1/3, 1/2)$ , exact arithmetic over  $\mathbb{Q}$  shows that  $\rho_{\tau_0}$  is injective on the ball of radius 10 and

that its only coincidences on the ball of radius 11 are the 33 listed pairs, each consisting of two words of length exactly 11; hence the layer counts are  $3 \cdot 2^{d-1}$  for  $1 \leq d \leq 10$  and  $3072 - 33 = 3039$  at  $d = 11$ . Any coincidence  $\rho_\tau(w) = \rho_\tau(w')$  with  $w \neq w'$  of length at most 11 is the vanishing of a rational function of  $(p, q)$ ; by the witness it is not identically zero unless  $(w, w')$  is one of the 33 pairs, so each of the finitely many remaining conditions defines a proper closed subset. Off their union the coincidences are exactly the 33 pairs.

Finally, no cancellation occurs in  $w_i(w'_i)^{-1}$ . If  $w_i$  and  $w'_i$  shared a common suffix, say  $w_i = us$  and  $w'_i = u's$ , then  $\rho_\tau(u) = \rho_\tau(u')$  with  $u \neq u'$  of length at most 10, contradicting injectivity on that ball; a common prefix is excluded in the same way. Hence each relator has length exactly 22, and by (ii) no shorter element of  $\ker \rho_\tau$  is common to all shapes.

For the radius 12 statement, enumeration of all reduced words of length at most 12 at the witness  $\tau_0$ , and independently at the witness with third vertex  $(2/7, 3/5)$ , returns the same 132 coincidences beyond those of the ball of radius 11 and no others; each of the 132 is an identity over  $\mathbb{Q}(p, q)$ , verified after clearing denominators as an identity of polynomials in  $\mathbb{Z}[p, q]$ . The argument of the previous paragraph applies verbatim to the finitely many remaining conditions on the ball of radius 12.  $\square$

*Remark 2.1.* The 132 coincidences at length 12 carry no new relations: 66 of them are the one letter extensions  $w_ix = w'_ix$  and  $xw_i = xw'_i$  of the 33 pairs, and 66 are the splittings of the 33 relators into a word of length 12 and a word of length 10, two per relator. In particular 66 reduced words of length 12 represent elements at distance 10, so the image  $\rho_\tau(\{|w| = 12\})$  has 6078 elements whereas the sphere of the word metric has 6012; it is the latter that the theorem records, and that the breadth-first computations return.

*Remark 2.2.* The witnesses  $(-1/3, 1/2)$ , obtuse, and  $(2/7, 3/5)$  produce the same 33 classes. By [1, Thm. 4.2] each of the 33 relators lies in the normal closure of the commutators of  $t_1$  with its conjugates, so the theorem is consistent with, and refines metrically, the presentation given there. Their Section 6 tabulates cyclically reduced stable words up to length 24 modulo symmetries; that census is of the relator language, not of the geodesic ball.

### 3 Sites, lamps and the hexagonal lattice

Normalize the incircle of  $\tau$  to be the unit circle at the origin, so that  $(x)r_i = (x)s_i + 2v_i$  with  $s_i$  the linear part and  $v_i$  the unit vector to the tangency

point of the  $i$ th side [1, §3]. Set  $t_{n,m} = ((t_1))(r_0 r_1)^n (r_0 r_2)^m$ ; by [1, Thm. 4.1] these form a basis of  $T_\tau$  for generic  $\tau$ . Conjugation by the generators acts on the index by

$$\sigma_0(n, m) = (-n-1, -m+1), \quad \sigma_1(n, m) = (-n-2, -m+1), \quad \sigma_2(n, m) = (-n-1, -m),$$

three affine involutions whose pairwise products are the translations by

$$a = (1, 0), \quad b = (0, 1), \quad c = (1, -1), \quad a = b + c.$$

We call  $\{\pm a, \pm b, \pm c\}$  the *hex generators* and say two sites are *hex-adjacent* if their difference is one of them; the sites  $(0, 0), (-1, 0), (-1, 1)$  are the *base sites*.

**Proposition 3.1** (Reduction). *Distinct reduced integer vectors on sites represent distinct elements of  $T_\tau$ . In particular no site carries both signs in a reduced representative, and a configuration containing a lamp together with its inverse at one site represents the identity.*

*Proof.* Immediate from the freeness of  $T_\tau$  on the conjugacy class of  $t_1$  [1, Thm. 4.1], together with injectivity of  $(n, m) \mapsto t_{n,m}$  for generic  $\tau$ .  $\square$

## 4 Single lamps

Let  $k(n, m)$  be the least length of a non-backtracking walk in the  $\sigma_i$  from a base site to  $(n, m)$  whose first step is by the middle letter of the cyclic word attached to that base site.

**Proposition 4.1.**  $\ell(t_{n,m}) \leq 6 + 2k(n, m)$ , with equality for every site with  $k \leq 5$ .

*Proof.* Conjugating a cyclic word  $(abc)^2$  by its head or tail letter is a rotation and preserves length; conjugating by the middle letter lengthens it by 2, after which the word begins and ends with that letter, so the only length preserving conjugation is the inverse step and every further conjugation by a different letter adds 2. A minimal walk therefore realizes  $t_{n,m}$  by a word of length  $6 + 2k(n, m)$ . Equality on the stated range is exact breadth-first search.  $\square$

Corollary 5.8 upgrades the inequality to an equality at every site, for flow-generic shapes.

## 5 The translation metric

Write  $F = C_2 * C_2 * C_2$  as before and let

$$\widehat{P} = \langle x_0, x_1, x_2 \mid x_i^2, [x_0x_1, x_0x_2] \rangle.$$

The rotations  $\rho_1 = x_0x_1$  and  $\rho_2 = x_0x_2$  commute in  $\widehat{P}$  and are inverted by conjugation with  $x_0$ , so every element has the normal form  $\rho_1^{m_1}\rho_2^{m_2}x_0^\epsilon$  and  $\widehat{P} \cong \mathbb{Z}^2 \rtimes C_2$ . Let  $X$  be the left Cayley graph of  $\widehat{P}$  on  $x_0, x_1, x_2$ , with edges  $\{g, x_i g\}$  and base vertex  $e$ : a connected cubic simple graph, bipartite for the orientation character  $\epsilon(x_i) = -1$ . For every nondegenerate  $\tau$  the linear parts satisfy  $s_i^2 = 1$  and  $[s_0s_1, s_0s_2] = 1$ , plane rotations commuting, so  $x_i \mapsto s_i$  defines a homomorphism  $M_\tau: \widehat{P} \rightarrow \mathrm{O}(2)$  at every shape.

### 5.1 Walks, flows, deposits

For a word  $w = x_{i_1} \cdots x_{i_\ell}$  let  $\sigma_j \in \widehat{P}$  be the image of the suffix  $x_{i_{j+1}} \cdots x_{i_\ell}$ , so that  $\sigma_\ell = e$  and  $\sigma_{j-1} = x_{i_j} \sigma_j$ . The *suffix walk*  $\sigma_\ell, \dots, \sigma_0$  traverses at step  $j$  the edge  $\{\sigma_j, x_{i_j} \sigma_j\}$  of  $X$ . Orient each edge of  $X$  from its  $\epsilon = +1$  endpoint and let the *flow*  $\phi(w) \in \mathbb{Z}^{E(X)}$  be the net signed traversal count of the suffix walk. When  $w$  maps to 1 in  $\widehat{P}$  the walk is closed and  $\phi(w)$  is a *cycle*: at each vertex the signed values of the three incident edges sum to zero. Flow is additive under concatenation of words with trivial image.

In the incircle expansion of §3 the translation part of  $\rho_\tau(w)$  is

$$t(w) = \sum_{j=1}^{\ell} 2v_{i_j} s_{i_{j+1}} \cdots s_{i_\ell} = \sum_{j=1}^{\ell} 2v_{i_j} M_\tau(\sigma_j).$$

Step  $j$  deposits  $2v_{i_j} M_\tau(\sigma_j)$ , and the deposits of the two directions of one edge are opposite:  $v_i s_i = -v_i$  gives  $2v_i M_\tau(x_i \sigma) = -2v_i M_\tau(\sigma)$ . Writing  $d_\tau(E)$  for the deposit of the canonical direction of the edge  $E$ ,

$$t(w) = V_\tau(\phi(w)), \quad V_\tau(\phi) = \sum_{E \in E(X)} \phi_E d_\tau(E), \quad (1)$$

for every shape and every word with trivial image.

### 5.2 Faces

Let  $w_6 = (x_0x_1x_2)^2$ , the word of  $t_1$ .

**Lemma 5.1.** *Modulo the relations  $x_i^2$  one has  $w_6 = [\rho_1, \rho_2^{-1}]$ ; consequently  $\ker(F \rightarrow \hat{P})$  is the normal closure of  $w_6$ , and the flows of its elements form the  $\mathbb{Z}$ -span of the flows of the conjugates  $uw_6^{\pm 1}u^{-1}$ .*

*Proof.*  $x_0x_1x_2x_0x_1x_2 = (x_0x_1)(x_2x_0)(x_1x_0)(x_0x_2) = \rho_1\rho_2^{-1}\rho_1^{-1}\rho_2$ . Hence imposing  $w_6 = 1$  on the free product imposes exactly the commutation of  $\rho_1$  and  $\rho_2$ , which presents  $\hat{P}$ . A product of conjugates of  $w_6^{\pm 1}$  is a concatenation of words with trivial image, on which flow is additive.  $\square$

**Lemma 5.2** (Faces). *The support of  $\phi(w_6)$  is a hexagonal cycle through  $e$ , and  $\phi(uw_6^{\pm 1}u^{-1}) = \pm \phi(w_6) \cdot g^{-1}$ , where  $g$  is the image of  $u$  and  $\hat{P}$  acts on  $X$  by right translation. The translated hexagons, the faces of  $X$ , are indexed by the sites of §3 through the dictionary  $t_{n,m}$ ; write  $\partial f_s$  for the oriented flow of the face at site  $s$ . Then:*

- (i) *every vertex of  $X$  lies on exactly three faces, the faces at  $e$  being those of the three base sites, and every edge lies on exactly two;*
- (ii) *two faces share an edge exactly when their sites are hex-adjacent, and then they share one edge, with opposite orientations;*
- (iii) *the  $\partial f_s$  are linearly independent, and the flows of the elements of  $\ker(F \rightarrow \hat{P})$  are precisely the finite sums  $\sum_s c_s \partial f_s$  with  $c_s \in \mathbb{Z}$ .*

*Proof.* In the suffix walk of  $uw_6^{\pm 1}u^{-1}$  the suffixes of the  $u^{-1}$  segment retrace those of the  $u$  segment in the opposite direction, so their traversals cancel, while the suffixes of the middle segment are those of  $w_6^{\pm 1}$  multiplied on the right by  $g^{-1}$ ; this is the displayed formula. The suffix walk of  $w_6$  visits six distinct elements of  $\hat{P}$  and closes, giving a hexagon through  $e$ . Conjugation permutes the lamps through the involutions  $\sigma_i$  of §3, which indexes the orbit of hexagons by the sites. Right translation is a graph automorphism acting transitively on vertices, so (i) follows from its validity at  $e$ , where exact computation shows the three faces through  $e$  are those of the base sites  $(0,0), (-1,0), (-1,1)$  and each of the three edges at  $e$  lies on two of them. For (ii), faces sharing an edge have intersecting vertex sets, which confines the difference of their sites to a finite set; exact computation over this set shows a common edge occurs precisely at the six hex differences, is then unique, and carries opposite orientations in the two faces. For (iii), spanning is Lemma 5.1. Independence: order sites lexicographically and let  $s_0$  be maximal in the support of  $\sum_s c_s \partial f_s$ ; the edge shared by  $f_{s_0}$  and  $f_{s_0+a}$  receives  $\pm c_{s_0} \neq 0$ , since  $c_{s_0+a} = 0$ .  $\square$

### 5.3 Separation and genericity

**Lemma 5.3** (Separation). *For every nonzero finitely supported cycle  $\phi$  on  $X$ , the set of shapes  $(p, q)$  with  $V_\tau(\phi) = 0$  is a proper Zariski-closed subset of the shape plane.*

*Proof.* The entries of  $V_\tau(\phi)$  are rational functions of  $(p, q)$ , so it suffices to show that  $V_\tau(\phi)$  does not vanish identically. Every edge  $E$  has one endpoint with  $\epsilon = +1$ ; write its normal form  $\rho_1^{m_1}\rho_2^{m_2}$  and let  $i$  be the letter of  $E$ . Let  $\beta_1, \beta_2$  be the rotation angles of  $M_\tau(\rho_1), M_\tau(\rho_2)$  and  $\theta_i$  the direction of the  $i$ th side, so that  $\theta_1 \equiv \theta_0 - \beta_1/2$  and  $\theta_2 \equiv \theta_0 - \beta_2/2$  modulo  $\pi$ , up to one global choice of sign of the  $\beta_k$ . The deposit  $d_\tau(E)$  is a nonzero vector in the direction  $\theta_i + \frac{\pi}{2} + \eta(m_1\beta_1 + m_2\beta_2)$  modulo  $\pi$ , with  $\eta = \pm 1$  one global sign. The frequency of  $E$ , the coefficient vector of  $(\beta_1, \beta_2)$  in this direction, lies in  $(\frac{1}{2}\mathbb{Z})^2$ ; doubled, the frequencies of edges with letter 0, 1, 2 are congruent to  $(0, 0), (1, 0), (0, 1)$  modulo 2 respectively. With the two global signs fixed, the letter of  $E$  is determined by the parity of its doubled frequency and the exponents  $(m_1, m_2)$  by its integer part, so distinct edges have distinct frequencies. If  $V_\tau(\phi)$  vanished identically then, the map from shapes to  $(\beta_1, \beta_2)$  being open near a suitable shape, a finite sum  $\sum_E \pm \phi_E |d_\tau(E)| e^{i(c_E + f_E \cdot \beta)}$  with distinct frequencies  $f_E$  would vanish on a nonempty open set of  $\beta$ ; substituting  $z_k = e^{i\beta_k/2}$  makes it a Laurent polynomial vanishing on an open subset of the torus, so all coefficients vanish, and  $|d_\tau(E)| \neq 0$  forces  $\phi = 0$ .  $\square$

**Definition 5.4.** The shape  $\tau$  is *flow-generic* if  $M_\tau$  is injective and  $V_\tau(\phi) \neq 0$  for every nonzero finitely supported cycle  $\phi$  on  $X$ .

**Proposition 5.5.** *Flow-genericity fails only on a countable union of proper Zariski-closed subsets of the shape plane. In particular every  $\tau$  whose coordinates  $p, q$  are algebraically independent over  $\mathbb{Q}$  is flow-generic.*

*Proof.* Injectivity of  $M_\tau$  reduces to injectivity on the rotation subgroup, reflections never being trivial, and there it is the condition  $m_1\beta_1 + m_2\beta_2 \notin 2\pi\mathbb{Z}$  for  $(m_1, m_2) \neq (0, 0)$ ; each failure is a nontrivial algebraic condition holding at some shapes and not at others, hence a proper Zariski-closed set, and there are countably many. With Lemma 5.3 this exhausts the conditions, each defined over  $\mathbb{Q}$ ; a point with algebraically independent coordinates lies on no proper  $\mathbb{Q}$ -defined Zariski-closed subset.  $\square$

**Theorem 5.6** (Structure). *Let  $\tau$  be flow-generic. Every word representing a translation has closed suffix walk; all words representing the same translation have the same flow; and  $c \mapsto \phi(c) := \sum_s c_s \partial f_s$  identifies the finitely*

supported integer vectors on sites both with the cycles of Lemma 5.2(iii) and with  $T_\tau$ . In particular  $T_\tau$  is free abelian on the lamps, recovering [1, Thm. 4.1] for flow-generic shapes, and

$$\|\phi(c)\|_1 = \sum_{\{u,v\}} |c_u - c_v|, \quad (2)$$

the sum over unordered hex-adjacent pairs of sites with  $c$  extended by zero.

*Proof.* A word  $w$  with  $\rho_\tau(w) \in T_\tau$  has trivial linear part, hence trivial image in  $\hat{P}$  by injectivity of  $M_\tau$ , and  $\phi(w)$  lies in the span of the faces by Lemma 5.2(iii). If  $w, w'$  represent the same translation then  $V_\tau(\phi(w) - \phi(w')) = 0$  by (1), so  $\phi(w) = \phi(w')$  by flow-genericity. Taking products of conjugated lamp words matches the flow of the configuration  $c$  with  $\sum_s c_s \partial f_s$ , injectivity of  $c \mapsto \phi(c)$  is Lemma 5.2(iii), and surjectivity onto  $T_\tau$  follows since every translation is represented by some word. For (2): every edge of the face of  $u$  is shared with exactly one neighbouring face  $v$  by Lemma 5.2(i), and carries the flow  $\pm(c_u - c_v)$  by (ii).  $\square$

## 5.4 The metric theorem

**Theorem 5.7** (Metric). *Let  $c$  be a nonzero reduced configuration with flow  $\phi = \phi(c)$ , and let  $\text{st}(c)$  be the least cardinality of a set  $M$  of edges of  $X$  outside the support of  $\phi$  such that the subgraph of  $X$  consisting of the edges  $\text{supp } \phi \cup M$ , their endpoints, and the vertex  $e$  is connected. Then*

$$\ell(c) = \|\phi(c)\|_1 + 2\text{st}(c)$$

for flow-generic  $\tau$ , and  $\ell(c) \leq \|\phi(c)\|_1 + 2\text{st}(c)$  for every  $\tau$ . In particular  $\ell(c) \equiv \|\phi(c)\|_1 \pmod{2}$ , and every translation has even length.

*Proof.* Lower bound. Let  $w$  be any word representing  $c$  and let  $n_E$  count all traversals of the edge  $E$  by its suffix walk, so that  $\ell(w) = \sum_E n_E$ , and, the two directions of an edge contributing opposite signs,  $n_E \geq |\phi_E|$  and  $n_E \equiv \phi_E \pmod{2}$ , where  $\phi = \phi(w) = \phi(c)$  by Theorem 5.6. The traversed edges form a connected subgraph containing  $e$ , consecutive steps sharing a vertex and the walk beginning at  $e$ . Let  $M$  be the set of traversed edges with  $\phi_E = 0$ ; each has  $n_E \geq 2$  by parity, the set  $\text{supp } \phi \cup M \cup \{e\}$  is connected, so  $|M| \geq \text{st}(c)$ , and

$$\ell(w) = \sum_{\text{supp } \phi} n_E + \sum_M n_E \geq \|\phi\|_1 + 2|M| \geq \|\phi\|_1 + 2\text{st}(c).$$

The congruence follows by summing  $n_E \equiv \phi_E \pmod{2}$ , and evenness from (2), each site having six incident lattice edges.

Upper bound. Let  $M$  attain  $\text{st}(c)$ . Form the directed multigraph with  $|\phi_E|$  parallel copies of each support edge, directed by the sign of  $\phi_E$ , and one copy of each direction of each edge of  $M$ . Every vertex is balanced: the cycle condition at a vertex is the balance of the signed values of its incident edges, and the doubled edges of  $M$  are balanced. The multigraph is connected, contains  $e$ , and  $e$  is incident to an edge, so it carries an Eulerian circuit through  $e$ ; reading the letters of its edges in reverse order yields a word whose suffix walk is the circuit, of length  $\|\phi\|_1 + 2|M|$ , with flow  $\phi$  and trivial image in  $\widehat{P}$ . Its linear part is trivial and its translation part is  $V_\tau(\phi(c))$  by (1), so it represents  $c$ , at every shape.  $\square$

**Corollary 5.8** (Single lamps). *For flow-generic  $\tau$ ,  $\ell(t_{n,m}^{\pm 1}) = 6 + 2d_X(e, f_{n,m})$ , the graph distance in  $X$  from the base vertex to the vertex set of the face of  $(n, m)$ . The values  $d_X(e, f_{n,m})$  agree with the walk count  $k(n, m)$  of §4 throughout the computed range  $|n|, |m| \leq 7$ .*

*Proof.* Here  $\|\phi\|_1 = 6$ . A geodesic from  $e$  to the nearest vertex of the face uses no face edge and connects, so  $\text{st} \leq d_X(e, f_{n,m})$ ; conversely the connected graph  $\text{supp } \phi \cup M \cup \{e\}$  contains a path from  $e$  to the face whose edges, up to its first face vertex, all lie in  $M$ , so  $\text{st} \geq d_X(e, f_{n,m})$ . The agreement with  $k$  is exact computation.  $\square$

**Corollary 5.9** (Cancellation).  $\|\phi(c)\|_1 = 6 \sum_s |c_s| - 2 \sum_{\{u,v\}} \min(|c_u|, |c_v|)$ , the second sum over hex-adjacent pairs of sites carrying equal signs. For configurations without repeated lamps this is  $6L - 2h$ , with  $L$  the number of lamps and  $h$  the number of same-sign hex-adjacent site pairs.

*Proof.* On the edge between adjacent sites  $u, v$  one has  $|c_u - c_v| = |c_u| + |c_v|$  when the signs differ or one side vanishes, and  $|c_u| + |c_v| - 2 \min(|c_u|, |c_v|)$  when they agree; sum over the six edges of each site and halve the double count.  $\square$

**Corollary 5.10** (Connected stratum).  $\text{st}(c) = 0$  if and only if the support of  $\phi(c)$  together with  $e$  is connected, and then  $\ell(c) = \sum_{\{u,v\}} |c_u - c_v|$ .

Two families illustrate the connectivity hypothesis. For the configuration with one lamp of a common sign at each of the three base sites, the three edges at  $e$  are the edges shared by the three base faces and their flows cancel, so  $e$  is isolated from the support:  $\text{st} = 1$  and  $\ell = 12 + 2 = 14$ . For the twelve sites at hex-distance two from a fixed site, each carrying one

lamp of the same sign, the support of the flow consists of two vertex-disjoint cycles, so  $\text{st} \geq 1$  for every translate of the configuration, although the site set is hex-connected; translated to contain a base site it has  $\text{st} = 1$  and  $\ell = 48 + 2 = 50$ . Connectivity of the site set therefore does not suffice, and the support condition of Corollary 5.10 cannot be weakened.

The number of translations of each word length for flow-generic  $\tau$  is, as computed below,

$$6, 6, 42, 96, 350, 1092, 3684$$

at lengths 6, 8, 10, 12, 14, 16, 18 respectively, and odd lengths carry none, as the theorem requires: enumeration of the configurations with  $\|\phi(c)\|_1 + 2\text{st}(c) \leq 18$ , over the bounded region described in the repository script, yields these counts, and exact rational breadth-first search at the witness with apex  $(2/7, 3/5)$  returns the same sets of translation vectors, element by element at every length, with distinct configurations receiving distinct vectors. At the witness with apex  $(1/3, 1/2)$  the same search returns 3680 at length 18: that shape is not flow-generic, and four coincidences mask the count there. The configurations with one positive lamp at each of  $(0, -2), (0, -1), (1, -2), (2, -2)$  and at each of  $(0, 1), (1, 1), (2, 0), (2, 1)$ , of length 18 at flow-generic shapes, represent the same translation at that witness, as do the configurations on  $(-2, -1), (-2, 0), (-1, -1), (0, -1)$  and on  $(-2, 2), (-1, 2), (0, 1), (0, 2)$ , together with the inverses of both pairs; at every other length the two witnesses agree. Of the 500 translations of length at most 14, the values  $\text{st} = 0, 1, 2, 3, 4$  occur 264, 152, 54, 12, 18 times.

## 6 Non-generic strata

**Proposition 6.1.** *The growth series of  $W_m = \langle x_0, x_1, x_2 \mid x_i^2, (x_0 x_1)^m \rangle$  with respect to  $x_0, x_1, x_2$  is*

$$W_m(t) = \frac{(1+t)(1+t+\dots+t^{m-1})}{1-t-t^2-\dots-t^m},$$

*so its growth rate is the  $m$ -bonacci constant.*

*Proof.* The parabolic formula for a Coxeter system with one finite rank two parabolic of order  $2m$  and two infinite bonds gives  $1/W_m(t^{-1}) = 1 - 3/(1+t) + 1/((1+t)(1+t+\dots+t^{m-1}))$ ; solving for  $W_m$  yields the displayed expression.  $\square$

Let one angle of  $\tau$  equal  $\pi/m$ , the shape being otherwise generic, and let  $d^*(m)$  be the first depth at which  $u_d$  falls below  $[t^d]W_m(t)$ , with deficit

$\delta(m)$  there. Exact computation in  $\mathbb{Q}(2\cos(\pi/m))$ , at two rational choices of the legs for each  $m$ , gives

| $m$      | 2  | 3 | 4 | 5  | 6 | 7  | 8  | 9  | 10 | 11 | 12 |
|----------|----|---|---|----|---|----|----|----|----|----|----|
| $d^*$    | 10 | 9 | 6 | 11 | 8 | 11 | 10 | 11 | 11 | 11 | 11 |
| $\delta$ | 8  | 3 | 3 | 33 | 8 | 33 | 15 | 33 | 33 | 33 | 33 |

The entries (11, 33) are the universal deficit of Theorem 1.1. Beyond the universal values, the deficit at depth 12 exceeds the generic 132 by 6 for  $m = 5$ , by 8 for  $m = 9$  and by 24 for  $m = 10$ , and equals it for  $m = 7, 11, 12$ .

The onset is therefore earlier than the universal depth 11 exactly for  $m \in \{2, 3, 4, 6, 8\}$  in this range. The crystallographic orders are not distinguished:  $m = 8$  has onset 10, and the mechanism is instead one of depth, as the following theorem shows.

**Theorem 6.2** (Rotation relations). *Let  $m \geq 3$  and let  $\tau$  have angle  $\pi/m$  at the vertex where the sides 0 and 1 meet, the shape being otherwise arbitrary. For a word  $w$  in  $r_0, r_1, r_2$  and a letter  $i$ , let  $c_i(w)$  be the number of occurrences of  $i$  at odd positions of  $w$  minus the number at even positions. Fix  $1 \leq c \leq m - 1$  and put  $n = m/\gcd(c, m)$ .*

- (i) *Let  $w$  have even length. Its linear part is, at every shape of the stratum, the rotation by  $2\pi c_1(w)/m$  if and only if  $c_2(w) = 0$ ; such a  $w$  represents a rotation of order  $m/\gcd(c_1(w), m)$  if  $m \nmid c_1(w)$ , and a translation otherwise.*
- (ii) *The reduced words of length  $2c + 2$  with  $c_2 = 0$  and  $c_1 = c$  are exactly the  $c^2$  words  $w_{a,b}$  with  $0 \leq a \leq c$ ,  $1 \leq b \leq c + 1$  and  $b \notin \{a, a + 1\}$ , in which every odd position carries 1 except position  $2a + 1$ , carrying 2, and every even position carries 0 except position  $2b$ , carrying 2. Reversal maps them bijectively onto the reduced words of length  $2c + 2$  with  $c_2 = 0$  and  $c_1 = -c$ , and no shorter word of even length has  $c_2 = 0$  and  $|c_1| = c$  except  $(r_0 r_1)^c$  and its reversal.*
- (iii) *Exactly one of the  $w_{a,b}$ , namely  $w_{0,c+1} = r_2(r_0 r_1)^c r_2$ , has finite order in  $W_m$ . Each of the other  $c^2 - 1$  satisfies  $w^n = 1$  throughout the stratum without this being a consequence of the Coxeter relations, and  $w^n$  is cyclically reduced of length  $n(2c + 2)$ , so splitting it in half exhibits a coincidence between two reduced words of length  $n(c + 1)$ ; they are distinct, since equality in  $F$  would give  $w^n = 1$  there. Whether the coincidence lowers the sphere at that depth requires the two halves to be geodesic in  $W_m$ ; this holds in the range  $c \leq m/2$  used in Corollary 6.3,*

where every  $\{0, 1\}$ -block of  $w_{a,b}$ , including the join  $u_2u_0$ , has length at most  $m$ , and can fail for larger  $c$ .

**Corollary 6.3.** *Let  $d$  be a divisor of  $m$  with  $2 \leq d \leq m/2$ . The stratum carries  $d^2 - 1$  relations at depth  $m + m/d$ . Taking  $d$  largest,  $d^*(m) \leq m + p$  with  $p$  the least prime factor of  $m$ , the family having  $(m/p)^2 - 1$  members. For  $m$  prime no such  $d$  exists, and minimising the depth  $(c+1)m/\gcd(c, m)$  of Theorem 6.2 over  $2 \leq c \leq m-1$  instead gives  $c = 2$ , so  $d^*(m) \leq 3m$  with 3 members. For even  $m$  this is  $d^*(m) \leq m + 2$  with  $(m/2)^2 - 1$  members, and the relations are half-turns,  $n = 2$  making each relation  $w = \bar{w}$ .*

*Proof.* (i) For even  $L$  the linear part of  $r_{a_1} \cdots r_{a_L}$  is the rotation by  $2 \sum_j (-1)^{j+1} \theta_{a_j} = 2(c_0\theta_0 + c_1\theta_1 + c_2\theta_2)$ , with  $\theta_i$  the direction of the  $i$ th side; normalize  $\theta_0 = 0$  and  $\theta_1 = \pi/m$ . As the shape runs through the stratum  $\theta_2$  runs through an interval while  $\theta_0, \theta_1$  stay fixed, so the linear part is the same at every shape of the stratum precisely when  $c_2 = 0$ , and then it is the rotation by  $2\pi c_1/m$ . A plane isometry whose linear part is a nontrivial rotation fixes a point and is the rotation by that angle about it, whence the stated order; if the linear part is trivial the isometry is a translation.

(ii) Put  $L = 2c + 2$  and let  $p, q$  count the letters 1 at odd and at even positions, so  $p - q = c_1 = c$ . Each parity class has  $c + 1$  positions, so  $p \leq c + 1$  and hence  $q \leq 1$ . If  $q = 1$  then  $p = c + 1$ , so every odd position carries 1; the even position carrying 1 has a neighbour, necessarily odd and carrying 1, contradicting reducedness. So  $q = 0$  and  $p = c$ : no even position carries 1, and exactly one odd position carries another letter. Were that letter 0, then no letter 2 would occur at an odd position, hence none at an even position either since  $c_2 = 0$ , so every even position would carry 0 and the exceptional odd position would have a neighbour carrying 0, again contradicting reducedness. Hence it carries 2, and  $c_2 = 0$  forces exactly one even position to carry 2, the others carrying 0. This is the displayed shape. Letters 2 at positions  $2a + 1$  and  $2b$  are adjacent exactly when  $b \in \{a, a + 1\}$ , and no other neighbouring letters coincide, so the admissible pairs are those listed,  $(c + 1)^2 - (2c + 1) = c^2$  in number. Reversal of a word of even length exchanges the two parity classes and so negates every  $c_i$ . For the last clause,  $|c_1| = c$  requires at least  $c$  letters 1 in one parity class, so an even length is at least  $2c$ ; at length  $2c$  the argument just given leaves only  $(r_0r_1)^c$  and its reversal. (For odd lengths the linear part is a reflection and (i) does not apply.)

(iii) By (i) each  $w_{a,b}$  is a rotation of order  $n$  at every shape of the stratum, so  $w_{a,b}^n = 1$  there. Since the Coxeter presentation of  $W_m$  imposes no relation

involving  $x_2$  beyond  $x_2^2$ ,

$$W_m \cong D_m * C_2, \quad D_m = \langle x_0, x_1 \rangle, \quad C_2 = \langle x_2 \rangle,$$

and by the torsion theorem for free products every element of finite order is conjugate into a factor; equivalently a cyclically reduced element with two or more syllables from  $D_m$  has infinite order. Write  $w_{a,b} = u_0 x_2 u_1 x_2 u_2$  with  $u_i \in D_m$  the images of its maximal  $\{0, 1\}$ -blocks. Its cyclic reduction is  $u_2 u_0 x_2 u_1 x_2$ , so,  $u_1$  being nontrivial by reducedness,  $w_{a,b}$  has finite order if and only if  $u_2 = u_0^{-1}$ . Suppose first  $b \geq a + 2$ . Reading the blocks gives  $u_0 = (x_1 x_0)^a$ ,  $u_1 = (x_0 x_1)^{b-a-1}$  and  $u_2 = (x_1 x_0)^{c+1-b}$ , so  $u_2 = u_0^{-1}$  amounts to  $b \equiv a + c + 1 \pmod{m}$ ; as  $2 \leq b - a \leq c + 1 \leq m$  this forces  $b - a = c + 1$ , that is  $a = 0$  and  $b = c + 1$ , where indeed  $w_{0,c+1} = r_2(r_0 r_1)^c r_2$  is conjugate to  $(r_0 r_1)^c$ . Suppose next  $b \leq a - 1$ . Then  $u_0 = (x_1 x_0)^{b-1} x_1$ ,  $u_1 = (x_1 x_0)^{a-b}$  and  $u_2 = x_0 (x_1 x_0)^{c-a}$ ; writing  $s = x_0 x_1$  and  $x_1 = x_0 s$ , the condition  $u_2 = u_0^{-1}$  becomes  $x_0 s^{a-c} = x_0 s^b$ , that is  $b \equiv a - c \pmod{m}$ , incompatible with  $1 \leq b \leq a - 1 \leq c - 1$ . Finally the first letter of  $w_{a,b}$  is 1 unless  $a = 0$  and the last is 0 unless  $b = c + 1$ , so for the remaining  $c^2 - 1$  words no cancellation occurs in  $w_{a,b}^n$ , which is therefore cyclically reduced of length  $n(2c + 2)$ .  $\square$

The counts 3, 8, 15, 24 at depths 6, 8, 10, 12 for  $m = 4, 6, 8, 10$  agree with Corollary 6.3;  $m = 2$  gives none, and  $m = 12$  is placed at depth 14 with 35 relations, beyond the computed window and consistent with the absence of extra relations at depth 12. For odd  $m$  the corollary gives 3 relations at depth 9 for  $m = 3$  and 8 at depth 12 for  $m = 9$ , both attained, and for the primes  $m = 5, 7, 11$  it gives 3 relations at depths 15, 21, 33. The bound is attained for every computed onset except two. For  $m = 2$  the theorem produces nothing, and for  $m = 5$  six relations occur already at depth 12, before the predicted 15. Of those six, three equate two translations: on the stratum  $(r_0 r_1)^m = 1$  identifies  $t_{n,j}$  with  $t_{n+m,j}$ , so the site lattice of §3 wraps into the cylinder  $(\mathbb{Z}/m) \times \mathbb{Z}$  and the flows of §5 live on a cylindrical honeycomb; the remaining three have  $c_2 = \pm 1$ . Theorem 6.5 below accounts for the translation kind: they are pairs of words whose flows differ by a wrapping cycle. We do not count them here, and for  $m = 7$  and  $m = 11$  we do not know whether relations of this second kind precede the depths 21 and 33 of the theorem. That the relations of Theorem 6.2 are the only coincidences at their depth, so that  $\delta$  equals the stated count rather than merely dominating it, is the computational input recorded above.

For  $m = 3, 4, 6, 8$  the relations tabulated above hold on the whole stratum: each is verified identically in the legs  $(a, b)$  over the relevant field, so

the onset is a property of the stratum and not of the samples, while the deficit being exactly  $\delta$  is witnessed at the rational samples and therefore holds off a proper closed subset of the stratum. The row  $m = 2$  is verified at three rational samples. The three relations at depth 9 for  $m = 3$  are

$$210201210 = 102012102, \quad 020121020 = 121020121, \quad 201210201 = 012102012,$$

of which the first and third have the form  $R s R s^{-1} R s$  with  $R = r_2$  and  $s = r_1 r_0$  the apex rotation of order three, while the second has as relator the cube  $(r_0 r_2 r_0 r_1 r_2 r_1)^3$ . The three at depth 6 for  $m = 4$  are

$$012102 = 201210, \quad 020121 = 121020, \quad 102012 = 210201.$$

## 6.1 The translation subgroup on a stratum

Let  $\tau$  be generic on the stratum  $\alpha = \pi/m$  and put  $\hat{P}_m = \hat{P}/\langle \rho_1^m \rangle \cong (\mathbb{Z}/m \times \mathbb{Z}) \rtimes C_2$ , with Cayley graph  $X_m$ : the quotient of the honeycomb  $X$  by the  $m$ -fold translation, a cylinder. Its faces are the wrapped sites  $(n \bmod m, j)$ , and the constructions of §5 apply to  $X_m$  unchanged, a word representing a translation having a closed suffix walk whose flow is a finitely supported cycle on  $X_m$  with  $t(w) = V_\tau(\phi(w))$ .

**Definition 6.4.** Call  $\tau$  *stratum-generic* for  $\alpha = \pi/m$  if  $M_\tau$  is injective on  $\hat{P}_m$ . Since  $M_\tau$  is injective on the rotation subgroup exactly when  $a(2\pi/m) + b\beta_2 \notin 2\pi\mathbb{Z}$  for  $(a \bmod m, b) \neq (0, 0)$ , and each failure is a nontrivial condition on the remaining shape parameter, this holds off countably many proper closed subsets of the stratum.

**Theorem 6.5** (Stratum metric). *Let  $\tau$  be stratum-generic for  $\alpha = \pi/m$ .*

- (i)  $\sum_{n \in \mathbb{Z}/m} t_{n,j} = 0$  for every  $j$ : the  $m$  lamps of a level sum to zero, so the wrapped lamps are not independent.
- (ii) The flow of a word representing a given translation is no longer unique. The word  $(r_0 r_1)^m$  and its conjugates have closed suffix walks, whose flows wrap once around the cylinder and are killed by  $V_\tau$ . Write  $\gamma_j$  for the flow of  $((r_0 r_1)^m)(r_0 r_2)^j$ , the conjugate by the  $j$ th power of  $r_0 r_2$ . Then  $\gamma_j - \gamma_{j-1}$  is the sum of the  $m$  faces of level  $j$ . Here  $Z_1(X_m)$  denotes the finitely supported cycles.
- (iii) For every translation  $g$ ,

$$\ell(g) = \min \{ \|\phi\|_1 + 2 \text{st}(\phi) : \phi \in Z_1(X_m), V_\tau(\phi) = g \}.$$

*Proof.* (i) Conjugation by  $r_0 r_1$  is the rotation by  $2\pi/m$  about the apex, so it rotates the translation vector of a lamp by that angle; the  $m$  vectors  $t_{n,j}$  with  $n \in \mathbb{Z}/m$  are therefore the images of one vector under the  $m$ th roots of unity, and those sum to zero. (ii)  $(r_0 r_1)^m$  is the rotation by  $2\pi$ , that is the identity, so the flow of its word has  $V_\tau = 0$ ; the identity between  $\gamma_j - \gamma_{j-1}$  and the level sum holds because both are the boundary of the annulus of  $X_m$  between the two wrapping curves. (iii) Neither bound of Theorem 5.7 used the planarity of  $X$  or the uniqueness of the flow. The lower bound gives  $\ell(w) \geq \|\phi(w)\|_1 + 2\text{st}(\phi(w))$  for every word  $w$  representing  $g$ , and  $\phi(w)$  is one of the competitors; conversely the Eulerian construction turns any competitor  $\phi$  into a word of length  $\|\phi\|_1 + 2\text{st}(\phi)$  with flow  $\phi$ , representing  $V_\tau(\phi) = g$ .  $\square$

For  $m = 5$  the formula was checked against exact rational breadth-first search on all 158 translations of length at most 12, the minimum being attained already among the flows differing from  $\phi(w)$  by a single  $\gamma_j$ . That stratum has 6, 6, 42, 104 translations at lengths 6, 8, 10, 12, against 6, 6, 42, 96 for a generic shape: wrapping shortens elements as well as identifying them.

The identifications themselves can be counted. Sending each of the 5276 generic translations of length at most 18 to its image on the stratum, the first coincidence appears at generic length  $2m + 4$  for  $m = 3, 5, 7$ , that is at 10, 14, 18. For  $m = 5$  and  $m = 7$  it is the plain wrap, the sites  $(2, 0)$  and  $(-3, 0)$  for  $m = 5$ , and  $(3, 0)$  and  $(-4, 0)$  for  $m = 7$ , which are identified by the cylinder and are the first such pair of equal generic length; for  $m = 3$  it is instead the level relation of part (i), which equates the lamp  $t_{0,0}$  of length 6 with the configuration  $-t_{-2,0} - t_{-1,0}$  of length 10. Within that window the numbers of identifications are 3255 for  $m = 3$ , 476 for  $m = 5$ , 14 for  $m = 7$ , and none for  $m = 9$  and  $m = 11$ ; for  $m = 3$  moreover 76 generic translations become trivial.

These identifications are of two kinds. Those between configurations with the same wrapped configuration hold already in  $W_m$ , where  $(x_0 x_1)^m = 1$  forces  $t_{n,j} = t_{n+m,j}$ ; they are 1438, 66 and 2 for  $m = 3, 5, 7$  and are not new relations. The remaining 1742, 410 and 12 come from the level relation of Theorem 6.5(i) and are new. The first of them is in each case one level split into two arcs,

$$t_{0,0} = -t_{1,0} - t_{2,0} \ (m = 3), \quad t_{3,0} + t_{4,0} = -\sum_{n \leq 2} t_{n,0} \ (m = 5), \quad \sum_{n \leq 2} t_{n,0} = -\sum_{n \geq 3} t_{n,0} \ (m = 7),$$

at generic lengths 6 and 10, 10 and 14, 14 and 18 respectively. An arc of  $k$  consecutive lamps of one sign has length  $4k + 2$ , by Corollary 5.9 and

Corollary 5.10, so the balanced split of a level of  $m$  lamps first becomes visible at depth  $4\lceil m/2 \rceil + 2$ , which is  $2m + 4$  for odd  $m$  and  $2m + 2$  for even  $m$ .

Comparing with Corollary 6.3: for even  $m$  the rotation relations at depth  $m + 2$  come first, and for odd composite  $m$  so does the divisor bound  $m + p$ ; for  $m = 3$  the rotation depth 9 precedes 10. But for every prime  $m \geq 5$  one has  $2m + 4 < 3m$ , so there the level relations arrive first, at depths 14, 18, 26 for  $m = 5, 7, 11$ . The stratum also carries translations that are not images of generic translations, a word being able to have trivial linear part on the stratum without having one generically. Both effects are accounted for by the following census.

**Proposition 6.6** (Stratum translation census). *For  $\tau$  generic on the stratum  $\alpha = \pi/m$ , the numbers of translations of each length are*

| <i>length</i>  | 6 | 8 | 10 | 12  | 14  |
|----------------|---|---|----|-----|-----|
| <i>generic</i> | 6 | 6 | 42 | 96  | 350 |
| $m = 3$        | 6 | 8 | 38 | 76  | 224 |
| $m = 5$        | 6 | 6 | 42 | 104 | 448 |
| $m = 7, 9, 11$ | 6 | 6 | 42 | 96  | 350 |

*In general the two censuses agree up to length  $2m + 1$  and differ at  $2m + 2$ .*

*Proof.* By Theorem 6.2(i) a word represents a translation on the stratum exactly when  $c_2 = 0$  and  $m \mid c_1$ . If  $c_1 = 0$  the linear part is trivial at every shape, so the word represents a generic translation as well. Otherwise  $|c_1| \geq m$ , so by Theorem 6.2(ii) the word has length at least  $2m$ , with equality only for  $(r_0 r_1)^{\pm m}$ , which is the identity on the stratum. Hence the shortest translation of the stratum that is not the image of a generic one has length  $2m + 2$ . Identifications among generic translations require the level relation, hence length at least  $2m + 4$  by the arc count above. Below  $2m + 2$  neither effect is available and the censuses agree; at  $2m + 2$  the first new translations appear. The displayed values are computed by exact rational breadth-first search in the stratum group at three leg samples for each  $m$ , at two leg samples for each  $m$ , which bounds the counts from below at any single sample and agrees at both; a configuration enumeration on  $X_m$  using the metric of Theorem 6.5 reproduces the same values in this range as a consistency check.  $\square$

For  $m = 5$  the stratum has *more* translations than a generic shape at the length where they first differ, the new translations outnumbering the

identifications; for  $m = 3$ , where the levels are short, the identifications dominate and the census falls below the generic one already at length 10. Beyond length 14 we do not report values: the configuration enumeration on  $X_m$  is not known to be exhaustive there, and for  $m = 9$  it undercounts, the correct value at length 18 being the generic 3684 by the argument of the proposition.

*Remark 6.7.* Rational leg samples that look generic need not be. For  $m = 3$  the legs  $(1, 2)$  give the right triangle with angles  $\pi/6, \pi/3, \pi/2$ , and for  $m = 5$  the legs  $(2, 3)$  produce 22 coincidences at length 14 that three other samples do not. Every computation reported here was repeated at three samples.

*Remark 6.8.* These lines lie outside the *typical* stratum of [1], where the angles are  $\mathbb{Q}$ -linearly independent and the shortest non-generic relations have length 18. On the lines  $\alpha = \pi/m$  the minimum drops as low as 12, attained by the relators of the three  $m = 4$  relations displayed above.

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**Data availability.** All code and verification data supporting the computational statements are openly available in the repository cited on the title page and archived on Zenodo (concept DOI 10.5281/zenodo.20370090).

**Use of AI tools.** Large language models (Anthropic Claude) were used in the preparation of this manuscript, both in drafting and in computation. Every computational claim is certified by the scripts in the cited repository, independently of any model output. The author takes full responsibility for the content.

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